AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Time series foundation models

Session 10 of 10





Outline

- What is a time series foundation model?
- How do they compare to classical methods?
- When to use a foundation model

What is a time series foundation model?

- Large, pre-trained models (e.g. Chronos, TimeGPT, Moirai) that forecast directly from historical data
- Trained once on huge, diverse collections of time series
- Little or no task-specific training required to forecast a new series (“zero-shot”)

How do they compare to classical methods?

- Classical methods (ETS, ARIMA, regression) are fit to *your* series
- Foundation models are pre-trained on *many* series, then applied to yours
- Trade-offs: convenience and speed vs. interpretability and control

When to use a foundation model

- Useful for quick baselines, and series with little historical data
- Classical methods often still competitive, and more transparent, once tuned
- Consider the operational context (compute, explainability, cost)

Summary

- Foundation models are a fast-moving area of forecasting research
- A useful addition to the toolkit alongside ETS, ARIMA and regression
- This session gives a high-level introduction; hands-on examples to follow